

TABLE I

ESTIMATION-AWARE CONTROL FOR AGILE QUADROTORS. $\hat{m}/\hat{\tau}$ = MASS / BODY-RATE LAG ESTIMATED ONLINE (✓ ONLINE, OFF = OFFLINE, ✗ NO); CL = CLOSED-LOOP CORRECTION OF PREDICTION MODEL; ADP = AGGRESSIVENESS MODULATED BY ESTIMATION CONFIDENCE.

Work	Ctrl	$\hat{m}$	$\hat{\tau}$	Dist.	CL	Adp	v [m/s]
[1] Kamel	NMPC	✗	✗	—	✗	✗	<5
[6] Torrente	NMPC	✗	✗	GP [†]	~	✗	14
[3] Romero	MPCC	✗	✗	—	✗	✗	16
[4] Krinner	MPCC	✗	✗	NN [†]	~	✗	>22
[5] Hanover	NMPC	✗	✗	$\mathcal{L}_1$	✗	✗	20
[7] Wang	Geom.	✗	✗	wrench	~	✗	<5
[9] Eschmann	NMPC	~	off	—	✗	✗	<5
[12] Gupta	NMPC	✗	off	—	✗	✗	27
[10] Fan	NMPC	~	✗	—	✓	✗	<5*
[11] Zhang	NMPC	✗	✗	MRAC	~	✗	<5
Ours	MPCC	✓	✓	MHE	✓	✓	10

[†]Offline learned. *Simulation only. Dist. column: estimation method for translational disturbance; all disturbances in acceleration [m/s²] except Wang (wrench [N, Nm]).

$\hat{w} \in \mathbb{R}^6$ and feed it back to the flight controller in real time, reducing force estimation error by up to 76.7% over neural baselines. A further extension, contextual NeuroMHE [13], improves generalisation across varying environmental conditions. Choo and Kayacan [8] combine data-driven MHE with Gaussian Processes for agile quadrotor flight, though hardware validation above 8 m/s has not been reported. Critically, none of these estimators isolate the body-rate controller lag τ_{rc} , which introduces systematic phase errors at high angular-rate commands that cannot be absorbed by an external-force estimate.

Offline low-level model identification. Eschmann *et al.* [9] identify motor delay time constants and inertia parameters via Maximum A Posteriori estimation from one minute of flight data. Gupta *et al.* [12] embed identified low-level controller dynamics, including τ_{rc} , directly into the NMPC prediction model, achieving speeds up to 98 km/h. Both works perform identification *offline*: τ_{rc} is assumed constant and is never re-estimated once airborne. Fan and Copp [10] propose an online moving-horizon parameter estimator feeding back structural and thrust parameters to NMPC, but validation is limited to simulation.

Identified Gap

As Table I summarises, each existing approach addresses only a subset of the three mismatch sources described above. Fan and Copp [10] estimate structural parameters online via moving-horizon estimation in closed loop with NMPC, but exclude actuator dynamics and external disturbances, and validate in simulation only. NeuroMHE [7] estimates the full external wrench online, yet does not isolate mass or body-rate lag as separate quantities and operates with a geometric controller at low speeds, not with a predictive controller. $\mathcal{L}_1$ -augmented NMPC [5] rejects disturbances at high bandwidth but treats them as a lumped scalar signal, leaving the prediction model itself uncorrected. Zhang *et al.* [11] cascade dual

MRAC channels onto NMPC to handle payload changes and impacts, but the adaptation is lumped and does not decompose into separate mass, actuator, and disturbance estimates. Offline identification [9], [12] captures the body-rate lag accurately but cannot track its variation in flight. Furthermore, all existing MPCC formulations [3], [4] use a fixed progress weight tuned offline; the speed–accuracy trade-off is never adapted to the current estimation confidence. To the best of our knowledge, the joint online estimation of mass, actuator lag, and translational disturbance acceleration — combined with closed-loop injection into MPCC and confidence-driven progress modulation — has not been addressed in the literature.

Contributions

- 1) An MHE that jointly estimates mass, body-rate lag, and translational disturbance acceleration online at 100 Hz, using a Lie-invariant arrival cost on the quaternion manifold.
- 2) Closed-loop injection of the MHE estimates into the MPCC prediction model, correcting mass, actuator dynamics, and disturbance at every control step.
- 3) An adaptive progress weight driven by the MHE posterior covariance, which automatically limits path speed when estimates are uncertain and releases it as they converge.
- 4) Validation in software-in-the-loop simulation under mass mismatch, constant wind, and gust disturbances, complemented by hardware experiments on a physical quadrotor.

The remainder of this paper is organised as follows. Section II introduces the quadrotor dynamics and path parameterisation. Sections III and IV formulate the MHE and the adaptive MPCC, respectively. Section V describes the closed-loop integration. Section VI presents SiL and hardware experiments, and Section VII concludes.

II. SYSTEM MODEL

A. Notation

Vectors are set in bold italic $\mathbf{a}$, matrices in bold upright $\mathbf{A}$. The inertial frame is $\mathcal{I}$ and the body frame is $\mathcal{B}$. We write $[\mathbf{a}]_{\times} \in \mathbb{R}^{3 \times 3}$ for the skew-symmetric matrix such that $[\mathbf{a}]_{\times} \mathbf{b} = \mathbf{a} \times \mathbf{b}$. The Hamilton quaternion product is denoted $\otimes$ and is defined so that $\mathbf{q}_1 \otimes \mathbf{q}_2$ corresponds to the rotation $\mathbf{R}(\mathbf{q}_1)\mathbf{R}(\mathbf{q}_2)$ (composition order). Unit quaternions are written $\mathbf{q} = [q_w, \mathbf{q}_v^T]^T \in \mathcal{S}^3 \subset \mathbb{R}^4$ with $\|\mathbf{q}\| = 1$. The weighted squared norm is $\|\mathbf{a}\|_{\mathbf{W}}^2 \triangleq \mathbf{a}^T \mathbf{W} \mathbf{a}$.

B. Quadrotor Physical Dynamics

The physical state is

$$\mathbf{x}_{\text{phy}} = [\mathbf{p}^T, \mathbf{v}^T, \mathbf{q}^T, \boldsymbol{\omega}^T]^T \in \mathbb{R}^3 \times \mathbb{R}^3 \times \mathcal{S}^3 \times \mathbb{R}^3, \quad (1)$$

where $\mathbf{p} \in \mathbb{R}^3$ is position in $\mathcal{I}$, $\mathbf{v} \in \mathbb{R}^3$ is inertial-frame velocity, $\mathbf{q} = [q_w, \mathbf{q}_v^T]^T \in \mathcal{S}^3$ is the unit quaternion mapping

$\mathcal{B} \rightarrow \mathcal{I}$, and $\boldsymbol{\omega} \in \mathbb{R}^3$ is the angular velocity in $\mathcal{B}$. The continuous-time dynamics are

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (2a)$$

$$\dot{\mathbf{v}} = -g \mathbf{e}_3 + \frac{f}{m} \mathbf{R}(\mathbf{q}) \mathbf{e}_3 + \mathbf{d}, \quad (2b)$$

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes [0, \boldsymbol{\omega}^\top]^\top, \quad (2c)$$

$$\dot{\boldsymbol{\omega}} = \frac{\boldsymbol{\omega}_c - \boldsymbol{\omega}}{\tau_{rc}}, \quad (2d)$$

where $g = 9.81 \text{ m/s}^2$, $\mathbf{e}_3 = [0, 0, 1]^\top$, m is the vehicle mass, $\boldsymbol{\omega}_c \in \mathbb{R}^3$ is the body-rate command, and $\tau_{rc} > 0$ is the first-order time constant of the low-level rate controller [9], [12]. In (2b), f denotes the *thrust state* (not a control input, see Section IV-A), and $\mathbf{d} \in \mathbb{R}^3$ is an acceleration disturbance (m/s^2 , inertial frame) capturing aerodynamic drag, payload shifts, and wind.

Implementation note. In the real-time OCP, equation (2d) is equivalently written as $\dot{\boldsymbol{\omega}} = k_\tau (\boldsymbol{\omega}_c - \boldsymbol{\omega})$ with the *rate gain* $k_\tau \triangleq 1/\tau_{rc}$ treated as a runtime parameter. This reparameterisation makes the dynamics *linear* in the estimated parameter, avoiding the $\mathcal{O}(1/\tau_{rc}^2)$ Jacobian entries that arise from differentiating $1/\tau_{rc}$ directly, which would otherwise ill-condition the HPIPM QP solver. The rotation matrix is

$$\mathbf{R}(\mathbf{q}) = (2q_w^2 - 1) \mathbf{I}_3 + 2q_w [\mathbf{q}_v]_\times + 2 \mathbf{q}_v \mathbf{q}_v^\top, \quad (3)$$

which implements the *active* body-to-inertial rotation under the Hamilton convention.

C. Path Parameterisation

A 3-D Lissajous figure is parameterised in arc-length θ via offline B-spline fitting to $N_{wp} = 400$ waypoints sampled from the time-parametric trajectory at $s_{\max} = 100 \text{ m}$ [3]. The B-spline map

$$\gamma(\theta) = [\mathbf{p}_d(\theta), \mathbf{q}_d(\theta), \hat{\boldsymbol{\omega}}(\theta)] \quad (4)$$

yields: path position $\mathbf{p}_d \in \mathbb{R}^3$, attitude reference $\mathbf{q}_d \in \mathcal{S}^3$ (from differential flatness [2], max tilt 60°), and the unit tangent $\boldsymbol{\tau}(\theta) = \partial \mathbf{p}_d / \partial \theta / \|\partial \mathbf{p}_d / \partial \theta\|$. A feedforward thrust could be derived from the path via differential flatness, but is unnecessary here because thrust is a state governed by the rate input Δf . The angular-velocity sensitivity $\hat{\boldsymbol{\omega}}(\theta) \in \mathbb{R}^3$ is the body-frame angular rate per unit arc-length, $\hat{\boldsymbol{\omega}} = 2(\mathbf{q}_d^* \otimes d\mathbf{q}_d/d\theta)|_{\text{vec}}$. The angular-velocity feedforward then scales with progress speed:

$$\boldsymbol{\omega}_d(\theta, v_\theta) = \hat{\boldsymbol{\omega}}(\theta) v_\theta. \quad (5)$$

III. LIE-INVARIANT MOVING HORIZON ESTIMATOR

A. Augmented State for Online Estimation

The uncertain parameters m , τ_{rc} , and $\mathbf{d}$ are treated as slowly varying states. Together with the physical state (1) and the MPCC virtual states (θ, v_θ, f) introduced in Section IV-A, the MHE augmented state is

$$\mathbf{x}_a = [\mathbf{x}_{\text{phy}}^\top, \theta, v_\theta, f, m, \tau_{rc}, \mathbf{d}^\top]^\top \in \mathbb{R}^{21}, \quad (6)$$

with random-walk drift dynamics

$$\dot{m} = w_m, \quad \dot{\tau}_{rc} = w_\tau, \quad \dot{\mathbf{d}} = \mathbf{w}_d, \quad (7)$$

where $\mathbf{w}_a = [w_m, w_\tau, \mathbf{w}_d^\top]^\top \in \mathbb{R}^5 \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_w)$.

B. Measurement Model

The odometry system provides $\mathbf{x}_{\text{phy}}$ at discrete times $t_k = k\Delta t$. Because the quaternion lives on $\mathcal{S}^3$, the measurement function is *nonlinear*:

$$\mathbf{y}_k = h(\mathbf{x}_{a,k}) + \mathbf{n}_k, \quad \mathbf{n}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_n), \quad (8)$$

where the residual $\mathbf{r}_k = \mathbf{y}_k - h(\mathbf{x}_{a,k}) \in \mathbb{R}^{12}$ is computed using the same Lie-invariant Log map as the arrival cost (Section III-D):

$$h(\mathbf{x}_a) = [\mathbf{p}^\top, \mathbf{v}^\top, \boldsymbol{\varepsilon}_q(\mathbf{q}, \mathbf{y}_q)^\top, \boldsymbol{\omega}^\top]^\top, \quad (9)$$

so the measurement cost becomes $\ell_y = \|\mathbf{r}_k\|_{\mathbf{R}_n^{-1}}^2$ with $\mathbf{R}_n = \text{diag}(\sigma_p^2 \mathbf{I}_3, \sigma_v^2 \mathbf{I}_3, \sigma_q^2 \mathbf{I}_3, \sigma_\omega^2 \mathbf{I}_3) \in \mathbb{R}^{12 \times 12}$. The quaternion block contributes a 3-dimensional tangent-space residual rather than the 4-dimensional raw difference. The parameters $(m, \tau_{rc}, \mathbf{d})$ are not directly measured; they are identified through prediction residuals in the dynamics (2).

C. MHE Optimal Control Problem

At time step k the MHE estimates the augmented state trajectory over the past N_e steps. Let $\bar{\mathbf{x}}_a$ and $\mathbf{P}$ denote the prior mean and covariance at the window start $k - N_e$. The estimator solves

$$\min_{\substack{\mathbf{x}_{a,j}, \mathbf{w}_{a,j} \\ j \in [k - N_e, k]}} \Phi_{\text{arr}}(\mathbf{x}_{a,k - N_e}) + \sum_{j=k - N_e}^{k-1} \left[\ell_y(\mathbf{y}_j, \mathbf{x}_{a,j}) + \|\mathbf{w}_{a,j}\|_{\mathbf{Q}_w^{-1}}^2 \right] + \ell_y(\mathbf{y}_k) \quad (10a)$$

$$\text{s.t. } \mathbf{x}_{a,j+1} = F(\mathbf{x}_{a,j}, \mathbf{u}_j) + \mathbf{G} \mathbf{w}_{a,j}, \quad (10b)$$

$$\mathbf{x}_{a,j} \in \mathcal{X}_a, \quad (10c)$$

where $F : \mathbb{R}^{21} \times \mathbb{R}^5 \rightarrow \mathbb{R}^{21}$ is the discretised augmented dynamics (2)–(7), $\mathbf{G} \in \mathbb{R}^{21 \times 5}$ maps process noise into the five parameter slots $[m, \tau_{rc}, \mathbf{d}^\top]^\top$, and $\mathcal{X}_a$ enforces physical bounds on the estimated parameters and states (see Table II). The measurement cost uses the Lie-invariant residual (9):

$$\ell_y(\mathbf{y}_j, \mathbf{x}_{a,j}) = \|\mathbf{y}_j - h(\mathbf{x}_{a,j})\|_{\mathbf{R}_n^{-1}}^2. \quad (11)$$

D. Lie-Invariant Arrival Cost

The arrival cost Φ_{arr} summarises information accumulated before the current window. A Euclidean formulation,

$$\Phi_{\text{arr}}^{\text{Eucl}}(\mathbf{x}_a) = \|\mathbf{x}_a - \bar{\mathbf{x}}_a\|_{\mathbf{P}^{-1}}^2, \quad (12)$$

treats quaternions as elements of $\mathbb{R}^4$. Because both $\mathbf{q}$ and $-\mathbf{q}$ represent the same rotation in $SO(3)$, this cost is *not* invariant under the double-cover map $\pi : \mathcal{S}^3 \rightarrow SO(3)$: when the estimator state crosses a hemisphere boundary, $\|\mathbf{q} - \bar{\mathbf{q}}\|^2$ reports a spurious error even though $\mathbf{R}(\mathbf{q}) = \mathbf{R}(\bar{\mathbf{q}})$.

We replace the quaternion block with the left-invariant Riemannian metric on $SO(3)$. Define the quaternion error

$$\mathbf{q}_e = \bar{\mathbf{q}}^{-1} \otimes \mathbf{q}, \quad \bar{\mathbf{q}}^{-1} \triangleq [\bar{q}_w, -\bar{\mathbf{q}}_v^\top]^\top, \quad (13)$$

and apply the hemisphere correction

$$\tilde{\mathbf{q}}_e = \text{sgn}(q_{e,w}) \mathbf{q}_e \quad (14)$$

to ensure $\tilde{q}_{e,w} \geq 0$, where $\text{sgn}(0) \triangleq 1$. The rotation-vector error in $\mathfrak{so}(3) \cong \mathbb{R}^3$ is

$$\epsilon_q(\mathbf{q}, \bar{\mathbf{q}}) = \text{Log}_{S^3}(\tilde{\mathbf{q}}_e) \triangleq 2 \frac{\text{atan2}(\|\tilde{\mathbf{q}}_{e,v}\|, \tilde{q}_{e,w})}{\|\tilde{\mathbf{q}}_{e,v}\| + \epsilon} \tilde{\mathbf{q}}_{e,v}, \quad (15)$$

with $\epsilon = 10^{-9}$ for numerical stability. The factor of 2 arises because the unit quaternion encodes the half-angle: $\mathbf{q} = [\cos(\phi/2), \sin(\phi/2) \mathbf{n}^\top]^\top$ for rotation angle ϕ about axis $\mathbf{n}$, so the standard $SO(3)$ logarithm $\text{Log}_{SO(3)}(\mathbf{R}) = \phi \mathbf{n}$ requires $\phi = 2 \arctan(\|\mathbf{q}_v\|, q_w)$.

The Lie-invariant arrival cost is

$$\Phi_{\text{arr}}(\mathbf{x}_a) = \|\mathbf{p} - \bar{\mathbf{p}}\|_{\mathbf{P}_p^{-1}}^2 + \|\mathbf{v} - \bar{\mathbf{v}}\|_{\mathbf{P}_v^{-1}}^2 + \|\epsilon_q\|_{\mathbf{P}_q^{-1}}^2 + \|\boldsymbol{\omega} - \bar{\boldsymbol{\omega}}\|_{\mathbf{P}_\omega^{-1}}^2 + \|\boldsymbol{\theta} - \bar{\boldsymbol{\theta}}\|_{\mathbf{P}_\theta^{-1}}^2, \quad (16)$$

where $\boldsymbol{\theta} \triangleq [m, \tau_{\text{rc}}, \mathbf{d}^\top]^\top \in \mathbb{R}^5$ collects the estimated parameters, $\mathbf{P}_{(\cdot)} \succ 0$ are the corresponding diagonal blocks of $\bar{\mathbf{P}}$, and the virtual-state block $\mathbf{P}_\xi \in \mathbb{R}^{3 \times 3}$ covers (θ, v_θ, f) . The quaternion block $\mathbf{P}_q \in \mathbb{R}^{3 \times 3}$ operates in the tangent space $\mathfrak{so}(3) \cong \mathbb{R}^3$, not $\mathbb{R}^4$, so $\bar{\mathbf{P}}$ is 20×20 rather than 21×21 .

The hemisphere correction (14) ensures that the arrival cost (16) is invariant under the double-cover sign flip $\mathbf{q} \mapsto -\mathbf{q}$: flipping the sign of $\mathbf{q}$ negates the error quaternion $\mathbf{q}_e$, but the $\text{sgn}(\cdot)$ factor absorbs the extra minus sign, leaving $\tilde{\mathbf{q}}_e$ and therefore Φ_{arr} unchanged. This prevents fictitious attitude jumps when the quaternion crosses the $q_w = 0$ hemisphere boundary during aggressive manoeuvres — precisely the regime encountered at peak-curvature points of the Lissajous trajectory (Section VI).

E. Arrival Cost Propagation

After each solved window, the prior is propagated by one step using an extended Kalman prediction:

$$\hat{\mathbf{x}}_a^+ = F(\hat{\mathbf{x}}_{a,k-N_e}, \mathbf{u}_{k-N_e}), \quad (17a)$$

$$\bar{\mathbf{P}}^+ = \mathbf{A}_k \bar{\mathbf{P}} \mathbf{A}_k^\top + \mathbf{G} \mathbf{Q}_w \mathbf{G}^\top, \quad (17b)$$

where $\mathbf{A}_k = \partial F / \partial \mathbf{x}_a|_{\hat{\mathbf{x}}_{a,k-N_e}}$ is the discrete-dynamics Jacobian and $\hat{\mathbf{x}}_{a,k-N_e}$ is the MHE solution at the oldest node. The Jacobian $\mathbf{A}_k$ is 20×20 because the quaternion block is evaluated in the $\mathbb{R}^3$ tangent space via $\delta \mathbf{q} \approx \frac{1}{2} \mathbf{q} \otimes [0, \delta \phi^\top]^\top$, consistent with the Log map (15).

IV. ADAPTIVE MODEL PREDICTIVE CONTOURING CONTROL

Given the parameter estimates $\hat{\boldsymbol{\theta}}_k$ and uncertainty σ_k from the MHE (Section III), we now formulate the MPCC that consumes these outputs to adapt its prediction model and aggressiveness online.

A. MPCC Virtual States and Control Input

Model Predictive Contouring Control appends three virtual states to track path progress [3]: arc-length $\theta \in \mathbb{R}_{\geq 0}$ [m], progress velocity $v_\theta \in \mathbb{R}_{\geq 0}$ [m/s], and collective thrust $f \in \mathbb{R}_{\geq 0}$ [N]. The *thrust-as-state* formulation [3] avoids discontinuous thrust commands: the control input is the thrust

rate $\Delta f = \dot{f}$, so f evolves smoothly. The full MPCC state and control vectors are

$$\mathbf{x} = [\mathbf{x}_{\text{phy}}^\top, \theta, v_\theta, f]^\top \in \mathbb{R}^{16}, \quad \mathbf{u} = [\Delta f, \boldsymbol{\omega}_c^\top, a_\theta]^\top \in \mathbb{R}^5, \quad (18)$$

where $a_\theta = \dot{v}_\theta$ is the progress acceleration. The virtual-state dynamics extend (2) with

$$\dot{\theta} = v_\theta, \quad \dot{v}_\theta = a_\theta, \quad \dot{f} = \Delta f. \quad (19)$$

All 16 states are integrated inside the OCP via ERK4; only the physical states $\mathbf{x}_{\text{phy}}$ are fed back from odometry at runtime (Section V).

B. Contouring and Lag Error Decomposition

At arc-length θ , the position error to the path is $\mathbf{e}_t(\theta) = \mathbf{p}_d(\theta) - \mathbf{p}$. This decomposes orthogonally into:

$$\mathbf{e}_l(\theta) = \boldsymbol{\tau}(\theta)^\top \mathbf{e}_t(\theta), \quad (20a)$$

$$\mathbf{e}_c(\theta) = \mathbf{P}_\perp(\theta) \mathbf{e}_t(\theta), \quad (20b)$$

where $\mathbf{P}_\perp(\theta) = \mathbf{I}_3 - \boldsymbol{\tau} \boldsymbol{\tau}^\top$ is the contouring projector. The decomposition satisfies $\|\mathbf{e}_t\|^2 = \mathbf{e}_l^2 + \mathbf{e}_c^2$.

C. Contouring Cost and Lie-Invariant Attitude Error

At prediction stage $j \in \{0, \dots, N\}$, with path parameter θ_j and drone attitude $\mathbf{q}_j$, the attitude error is computed via the same Log map used in Section III-D, evaluated at the path-referenced attitude $\mathbf{q}_d(\theta_j)$:

$$\epsilon_{q,j} = \text{Log}_{S^3}(\tilde{\mathbf{q}}_{e,j}), \quad \mathbf{q}_{e,j} = \mathbf{q}_d(\theta_j)^{-1} \otimes \mathbf{q}_j, \quad (21)$$

with hemisphere correction (14). Using the same geometric error in both the MHE arrival cost and the MPCC stage cost ensures end-to-end Lie-invariant consistency.

The angular-rate tracking error with feedforward (5) is

$$\tilde{\boldsymbol{\omega}}_j = \boldsymbol{\omega}_{c,j} - \hat{\boldsymbol{\omega}}(\theta_j) v_{\theta,j} \in \mathbb{R}^3. \quad (22)$$

The stage cost is

$$\ell(\mathbf{x}_j, \mathbf{u}_j) = \|\mathbf{e}_{c,j}\|_{\mathbf{Q}_c}^2 + e_{l,j}^2 Q_l + \|\epsilon_{q,j}\|_{\mathbf{Q}_q}^2 + W_{\Delta f} \Delta f_j^2 + \|\tilde{\boldsymbol{\omega}}_j\|_{\mathbf{W}_\omega}^2 - W_s(k) v_{\theta,j}, \quad (23)$$

and the terminal cost (no progress term) is

$$\ell_N(\mathbf{x}_N) = \|\mathbf{e}_{c,N}\|_{\mathbf{Q}_c}^2 + e_{l,N}^2 Q_l + \|\epsilon_{q,N}\|_{\mathbf{Q}_q}^2. \quad (24)$$

The negative term $-W_s(k) v_{\theta,j}$ maximises progress speed along the path; all other terms penalise deviations. The weight matrices $\mathbf{Q}_c, Q_l, \mathbf{Q}_q, \mathbf{W}_\omega \succ 0$ and scalar $W_{\Delta f} > 0$ are fixed; only $W_s(k)$ varies online.

D. MPCC Optimal Control Problem

At step k the MPCC solves the OCP over the 16-state model (18)–(19):

$$\min_{\mathbf{u}_0, \dots, \mathbf{u}_{N-1}} \sum_{j=0}^{N-1} \ell(\mathbf{x}_j, \mathbf{u}_j) + \ell_N(\mathbf{x}_N) \quad (25a)$$

$$\text{s.t. } \mathbf{x}_0 = \hat{\mathbf{x}}_k, \quad (25b)$$

$$\mathbf{x}_{j+1} = F(\mathbf{x}_j, \mathbf{u}_j; \hat{m}_k, \hat{k}_{\tau,k}, \hat{\mathbf{d}}_k), \quad (25c)$$

$$0 \leq f_j \leq T_{\max}, |\Delta f_j| \leq \Delta f_{\max}, \quad (25d)$$

$$\|\omega_{c,j}\|_{\infty} \leq \omega_{\max}, 0 \leq v_{\theta,j} \leq v_{\theta,\max}, \quad (25e)$$

$$|e_{l,j}| \leq D_{\max} \quad (\text{soft, L2 penalty}). \quad (25f)$$

The dynamics F in (25c) integrate the full 16-state model using the MHE parameter estimates $\hat{\theta}_k = [\hat{m}_k, \hat{k}_{\tau,k}, \hat{\mathbf{d}}_k^T]^T \in \mathbb{R}^5$ as fixed runtime parameters within the prediction horizon, where $\hat{k}_{\tau,k} = 1/\hat{\tau}_{rc,k}$ (see implementation note in Section II). To prevent premature injection of unconverged estimates, the parameter update is gated: only when $\sigma_k < \sigma_{\text{thr}}$ does the MPCC receive $\hat{\theta}_k$; otherwise it falls back to nominal values $\theta_0 = [m_0, k_{\tau,0}, \mathbf{0}]^T$. Furthermore, an *exponential moving average* (EMA) is applied to the raw MHE output before injection, suppressing window-shift step changes without introducing estimator bias. Constraint (25f) prevents excessive lag accumulation. All design parameters are listed in Table II (Section VI-A).

E. Adaptive Progress Weight

The scalar $\sigma_k = \text{tr}(\bar{\mathbf{P}}_{\theta,k})$, where $\bar{\mathbf{P}}_{\theta,k} \in \mathbb{R}^{5 \times 5}$ is the parameter uncertainty block of the MHE prior covariance at step k , quantifies how well $\hat{\theta}_k$ is currently known. We define the *adaptive progress weight*

$$W_s(k) = \frac{W_{s,\max}}{1 + \alpha \sigma_k}, \quad (26)$$

where $W_{s,\max} > 0$ sets the maximum aggressiveness and $\alpha > 0$ controls sensitivity.

Physical interpretation. When σ_k is large (estimator uncertain, e.g. unknown payload just attached), $W_s(k) \rightarrow 0$ and the MPCC does not push v_{θ} hard — it tracks the path conservatively while the MHE gathers information. As $\hat{m}_k$ and $\hat{\mathbf{d}}_k$ converge, σ_k decreases and $W_s(k) \rightarrow W_{s,\max}$: the controller progressively recovers near-optimal lap time with no manual intervention.

Implementation. $W_s(k)$ is computed outside the OCP and injected as runtime parameter $p[13]$ via the acados parameter interface at each control step. This requires no structural change to the OCP — only a scalar update costs negligible computation.

V. CLOSED-LOOP ARCHITECTURE

Algorithm 1 summarises the online loop. The MHE and MPCC share no internal solver state; their coupling is one-directional: the MHE output $(\hat{\mathbf{x}}_k, \hat{\theta}_k)$ is injected into the MPCC as initial condition and model parameters. The additional signal $\bar{\mathbf{P}}_{\theta,k}$ feeds the adaptive weight (26) without altering the OCP structure. At step 3, the MHE solution

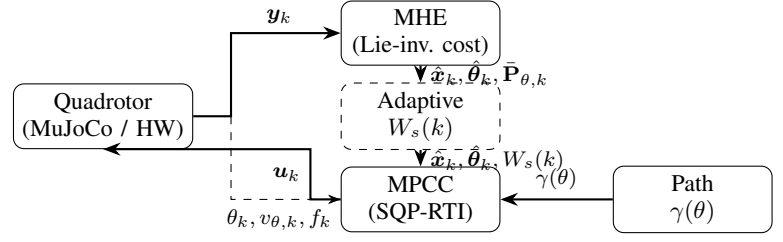


Fig. 1. Closed-loop MHE-MPCC architecture. The MHE accumulates $N_e = 20$ past measurements and returns: physical state estimate $\hat{\mathbf{x}}_k \in \mathbb{R}^{13}$, parameter estimate $\hat{\theta}_k = [\hat{m}_k, \hat{k}_{\tau,k}, \hat{\mathbf{d}}_k^T]^T \in \mathbb{R}^5$ (where $\hat{k}_{\tau} = 1/\hat{\tau}_{rc}$), and prior covariance block $\bar{\mathbf{P}}_{\theta,k} \in \mathbb{R}^{5 \times 5}$. The adaptive block computes $\sigma_k = \text{tr}(\bar{\mathbf{P}}_{\theta,k})$ and injects $W_s(k)$ as a runtime parameter into the MPCC. Virtual states $(\theta_k, v_{\theta,k}, f_k)$ are integrated outside the solver and fed forward to the MPCC initial condition. Both solvers run sequentially at 100 Hz.

Algorithm 1 Online MHE-MPCC control loop (100 Hz)

Require: Prior $(\bar{\mathbf{x}}_a, \bar{\mathbf{P}})$; precomputed path $\gamma(\theta)$; virtual states $(\theta_0, v_{\theta,0}, f_0)$.

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: Read odometry $\mathbf{y}_k$
- 3: **MHE:** solve (10) $\Rightarrow \hat{\mathbf{x}}_k \in \mathbb{R}^{13}, \hat{\theta}_k \in \mathbb{R}^5, \bar{\mathbf{P}}_{\theta,k} \in \mathbb{R}^{5 \times 5}$
- 4: Propagate arrival cost prior: $(\bar{\mathbf{x}}_a^+, \bar{\mathbf{P}}^+)$ via (17)
- 5: Compute uncertainty and adaptive weight: $\sigma_k \leftarrow \text{tr}(\bar{\mathbf{P}}_{\theta,k}), W_s(k) \leftarrow W_{s,\max}/(1 + \alpha \sigma_k)$
- 6: Apply EMA filter to raw MHE output: $\hat{\theta}_k^{\text{ema}} \leftarrow \alpha \odot \hat{\theta}_{k-1}^{\text{ema}} + (1 - \alpha) \odot \hat{\theta}_k$
- 7: Determine injected parameters: $\hat{\theta}_k^{\text{inj}} \leftarrow \hat{\theta}_k^{\text{ema}}$ if $\sigma_k < \sigma_{\text{thr}}$, else θ_0 (nominal)
- 8: **MPCC:** set initial condition $\mathbf{x}_0 = [\hat{\mathbf{x}}_k^T, \theta_k, v_{\theta,k}, f_k]^T$, update $\hat{\theta}_k^{\text{inj}}$ in (25c), inject $W_s(k)$ as runtime parameter; solve (25) $\Rightarrow \mathbf{u}_k^* = [\Delta f_k^*, \omega_{c,k}^*, a_{\theta,k}^*]^T$
- 9: Integrate virtual states (Euler): $\theta_{k+1} \leftarrow \theta_k + v_{\theta,k} \Delta t$; $v_{\theta,k+1} \leftarrow v_{\theta,k} + a_{\theta,k}^* \Delta t$; $f_{k+1} \leftarrow f_k + \Delta f_k^* \Delta t$
- 10: Apply commands: thrust f_k , body rates $\omega_{c,k}^*$
- 11: **end for**

$\hat{\mathbf{x}}_{a,k} \in \mathbb{R}^{21}$ is decomposed as: physical states $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{a,k}[1:13]$, virtual states $(\hat{\theta}_k, \hat{v}_{\theta,k}, \hat{f}_k) = \hat{\mathbf{x}}_{a,k}[14:16]$, and parameters $\hat{\theta}_k = [\hat{m}_k, \hat{\tau}_{rc,k}, \hat{\mathbf{d}}_k^T]^T = \hat{\mathbf{x}}_{a,k}[17:21]$, from which $\hat{k}_{\tau,k} = 1/\hat{\tau}_{rc,k}$ is computed before injection into the MPCC.

VI. EXPERIMENTAL RESULTS

This section validates the closed-loop MHE-MPCC architecture (Algorithm 1) in three stages: (i) SiL experiments with mass mismatch, (ii) SiL experiments with wind disturbances, and (iii) hardware experiments on a physical quadrotor.

A. Experimental Setup

1) *Platform:* The SiL experiments use MuJoCo with full rigid-body contact dynamics and a realistic motor model, interfaced via ROS 2 Humble at 100 Hz ($\Delta t = 10$ ms). Hardware experiments use the same software stack deployed on a custom quadrotor with onboard state estimation. Both solvers are generated with acados [14]/CasADi [15] and execute single-threaded on an Intel Core i7. Table II lists all design parameters.

TABLE II
MPCC AND MHE DESIGN PARAMETERS.

Parameter	MPCC	MHE
States / controls	16 / 5	21 (aug.) / 5
Horizon stages N	50	$N_e = 20$
Time grid T_h	1.5 s (non-unif.)	0.2 s
Integrator	ERK4	ERK4
QP solver	HPIPM	HPIPM
NLP solver	SQP-RTI (1 iter)	SQP-RTI (1 iter)
Regularisation	PROJECT + LM	PROJECT_REDUC + LM
m bounds	—	[0.5, 3.0 kg]
τ_c bounds	—	[0.005, 0.15 s]
$\ d\ _\infty$ bound	—	8 m/s ²
f bounds	[0, 49.05 N]	—
$\Delta f_{\max}$	500 N/s	—
$\omega_{\max}$	20 rad/s	—
a_θ bounds	$\pm 4g$ m/s ²	—
$v_{\theta, \max}$	14 m/s	—
$D_{\max}$ (lag)	0.5 m	—
σ_{thr}	0.05	—
$\alpha_{\text{EMA}, m/\tau}$	0.98	—
$\alpha_{\text{EMA}, d}$	0.92	—

2) *Path and speed*: The reference path is a three-dimensional Lissajous figure parameterised in arc-length with $s_{\max} = 100$ m (Section II-C), traversed at a maximum progress speed $v_{\theta, \max} = 10$ m/s.

3) *Mass mismatch scenario*: A deliberate mass error is introduced: the true vehicle mass is $m_{\text{true}} = 1.08$ kg while both modes are initialised with a nominal guess $m_0 = 0.9$ kg, corresponding to a -17% error.

4) *Wind disturbance scenarios*: Three wind profiles are applied via MuJoCo's external force interface to evaluate the disturbance estimator $\hat{d}$:

- **Constant wind**: a steady force ($\approx 10\%$ of vehicle weight) along the inertial x -axis throughout the flight.
- **Gust**: a short-duration impulse applied at mid-flight, simulating a localised wind burst.
- **Turbulence**: band-limited random forcing with a Dryden-like spectrum, applied on all three axes.

5) *Compared modes*: Two operating modes are compared:

- **v1 (open-loop baseline)**: The MHE runs but the injection gate is permanently disabled; the MPCC uses fixed nominal parameters $\theta_0 = [m_0, k_{\tau,0}, \mathbf{0}]^\top$.
- **v2 (closed-loop adaptive)**: The full architecture of Algorithm 1 is active: once $\sigma_k < \sigma_{\text{thr}}$, filtered estimates are injected into the MPCC dynamics (25c), and the adaptive progress weight (26) modulates $W_s(k)$ online.

B. SiL A: MHE Convergence under Mass Mismatch

Fig. 2 shows the evolution of the mass estimate $\hat{m}_k$, the parametric uncertainty σ_k , and the adaptive weight $W_s(k)$ during a representative v2 flight with 17% mass error.

C. SiL B: Ablation Study

To isolate the contribution of each component, four configurations are compared ($N \geq 10$ runs each, $v_{\theta, \max} = 10$ m/s, 17% mass error):

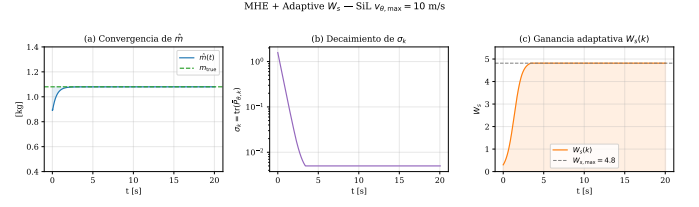


Fig. 2. MHE convergence during a v2 flight at $v_{\theta, \max} = 10$ m/s. **Top**: mass estimate $\hat{m}_k$ converging to m_{true} (dashed); the grey band marks the initial 17% error. **Bottom**: parametric uncertainty $\sigma_k = \text{tr}(\mathbf{P}_{\theta, k})$ (log scale, left axis) and adaptive progress weight $W_s(k)$ (right axis). The horizontal line indicates σ_{thr} .

TABLE III
ABLATION STUDY: POSITION RMSE (MEAN $\pm$ STD OVER N RUNS) AT $v_{\theta, \max} = 10$ m/s WITH 17% MASS ERROR.

Config	MHE $\hat{\theta} \rightarrow$ MPCC	Adp. W_s	RMSE [m]
A Baseline	✗	✗	□
B + adapt. W_s	✓	✗	□
C + param. inj.	✓	✓	□
D Full (ours)	✓	✓	□

- A** MPCC baseline — no MHE, fixed $W_s = W_{s, \max}$, nominal parameters.
- B** MHE runs, adaptive $W_s(k)$, but no parameter injection ($\hat{\theta}_k$ not sent to MPCC).
- C** MHE runs, parameter injection active, but $W_s = W_{s, \max}$ fixed.
- D** Full architecture (Algorithm 1): parameter injection *and* adaptive $W_s(k)$.

Table III reports the position RMSE for each configuration. [TODO: fill table and write analysis once $N \geq 10$ runs are collected for each configuration.]

D. SiL C: Tracking Performance

Fig. 3 shows the position error over arc-length for v1 (open-loop) and v2 (closed-loop adaptive).

Fig. 4 overlays the XY and XZ projections of both flights against the reference path.

E. SiL D: Wind Disturbance Rejection

Table IV compares v1 and v2 under the three wind profiles described in Section VI-A. [TODO: fill once wind experiments are executed.]

Fig. 5 shows the disturbance estimate $\hat{d}_k$ versus the true applied force (normalised by mass) for a representative constant-wind flight. [TODO: generate figure once wind experiments are executed.]

F. Hardware Experiments

The MHE–MPCC architecture is deployed on a physical quadrotor using the same software stack as in SiL (Section VI-A). A payload is attached to introduce a mass mismatch comparable to the SiL scenario.

Table V reports the tracking performance over $N \geq 5$ flights. [TODO: fill once hardware experiments are completed.]

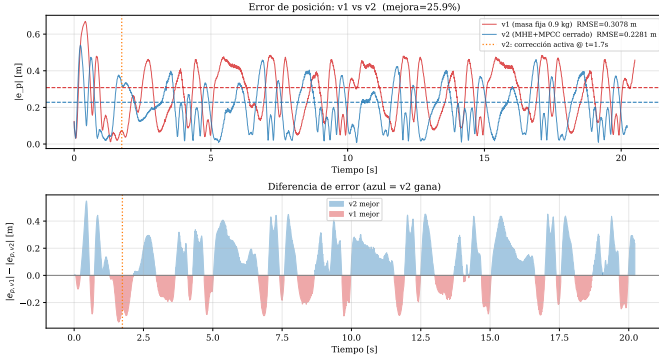


Fig. 3. Position error vs. arc-length θ for v1 (open-loop, dashed) and v2 (adaptive, solid). The shaded region highlights the error reduction after MHE convergence.

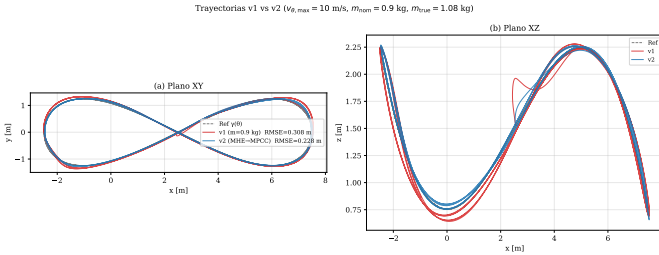


Fig. 4. XY and XZ trajectory projections for v1 (open-loop), v2 (adaptive), and the reference path.

Fig. 6 shows the recorded trajectory from a representative hardware flight. **[TODO: generate figure from hardware CSV data.]**

G. Computational Budget

Table VI reports the solver timing statistics. **[TODO: update timing values from final experimental data.]**

H. Discussion

[TODO: rewrite discussion once all experimental data (ablation, wind, hardware) are available. Key points to address: (1) ablation: which component contributes most; (2) wind: $\hat{d}$ captures perturbations in real time; (3) hardware: SiL results transfer to physical platform; (4) comparison with L1-adaptive approach.]

VII. CONCLUSION

We presented a closed-loop architecture coupling a Lie-invariant MHE with an adaptive MPCC for agile quadrotor path following under parametric uncertainty. The MHE jointly estimates mass, actuator lag, and translational disturbance acceleration at 100 Hz, and the posterior covariance drives an adaptive progress weight that modulates controller aggressiveness without manual intervention. SiL experiments with mass mismatch and wind disturbances, together with hardware flights on a physical quadrotor, demonstrate consistent tracking improvements over a fixed-model baseline without offline calibration or gain re-tuning. Future work will explore multi-run Monte Carlo campaigns across a wider range of flight

TABLE IV
TRACKING PERFORMANCE UNDER WIND DISTURBANCES
($v_{\theta, \max} = 10$ m/s, 17% MASS ERROR, MEAN $\pm$ STD OVER N RUNS).

Wind profile	v1 RMSE [m]	v2 RMSE [m]	Improv.
No wind (baseline)	☐	☐	☐
Constant	☐	☐	☐
Gust	☐	☐	☐
Turbulence	☐	☐	☐

Placeholder: $\hat{d}_k$ vs. true disturbance under constant wind.

Fig. 5. Disturbance estimation under constant wind. **Top:** true external acceleration (dashed) and MHE estimate $\hat{d}_k$ (solid). **Bottom:** estimation error $\|d - \hat{d}\|$.

speeds and disturbance magnitudes, and investigate code-generation optimisations to reduce the computational budget on embedded platforms.

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TABLE V
HARDWARE TRACKING PERFORMANCE ($v_{\theta, \max} = 10$ m/s,
PAYLOAD-INDUCED MASS MISMATCH, MEAN $\pm$ STD OVER N RUNS).

Metric	v1 (open-loop)	v2 (adaptive)
Position RMSE [m]	$\square$	$\square$
Max position error [m]	$\square$	$\square$
$\hat{m}$ converged [kg]	—	$\square$
Completion rate	$\square$	$\square$

Placeholder: hardware XY/XZ trajectory (v1 vs. v2 vs. reference).

Fig. 6. Hardware flight trajectory: v1 (open-loop), v2 (adaptive), and reference path.

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TABLE VI
SOLVER TIMING STATISTICS (INTEL CORE I7, SINGLE-THREADED).
BUDGET: 10 ms PER STEP AT 100 Hz.

Component	Mean [ms]	Peak [ms]
MPCC (SQP-RTI, $N = 50$)	$\square$	$\square$
MHE (SQP-RTI, $N_e = 20$)	$\square$	$\square$
MPCC + MHE combined	$\square$	—
Full loop	$\square$	$\square$

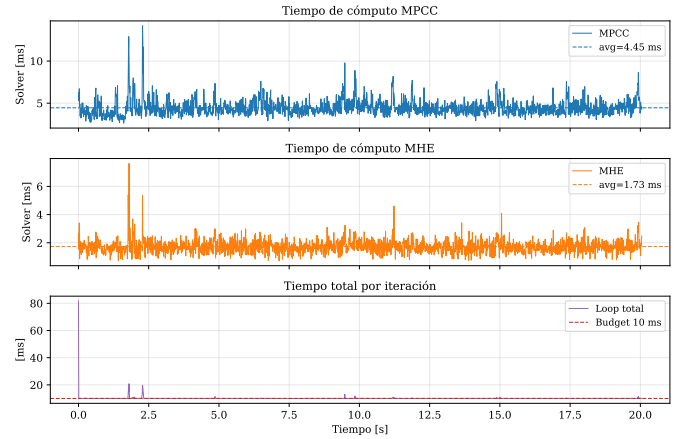


Fig. 7. Per-step solve time for the MPCC, MHE, and full control loop during a v2 flight. The dashed horizontal line marks the 10 ms real-time deadline.